



ICT272 WEB DESIGN AND DEVELOPMENT T126

All information in the Subject Outline is correct at the time of approval. KOI reserves the right to make changes to the Subject Outline if they become necessary. Any changes require the approval of the KOI Academic Board and will be formally advised to those students who may be affected by email and via Moodle.

Information contained within this Subject Outline applies to students enrolled in the trimester as indicated.

1. General Information

1.1 Administrative Details

Associated HE Award(s)	Duration	Level	Subject Coordinator
Bachelor of Information Technology (BIT)	1 trimester	Level 3	Umesh Poudel umesh.poudel@koi.edu.au P: +61 (2) 9283 3583 L: 7-11, 11 York St. Consultation: via Moodle or by appointment.

1.2 Core / Elective

Core subject for BIT

1.3 Subject Weighting

Indicated below is the weighting of this subject and the total course points.

Subject Credit Points	Total Course Credit Points
4	BIT (96 Credit Points)

1.4 Student Workload

Indicated below is the expected student workload per week for this subject

No. Timetabled Hours/Week*	No. Personal Study Hours/Week**	Total Workload Hours/Week***
4 hours/week (2 hour Lecture + 2 hour Tutorial)	6 hours/week	10 hours/week

* Total time spent per week at lectures and tutorials

** Total time students are expected to spend per week in studying, completing assignments, etc.

*** Combination of timetable hours and personal study.

1.5 Mode of Delivery Classes will be face-to-face or hybrid. Certain classes will be online (e.g., special arrangements).

1.6 Pre-requisites ICT104 Program Design and Development and ICT200 Database Design and Development;

1.7 General Study and Resource Requirements

- Dedicated computer laboratories are available for student use. Normally, tutorial classes are conducted in the computer laboratories.
- Students are expected to attend classes with the requisite textbook and must read specific chapters prior to each tutorial. This will allow them to actively take part in discussions. Students



should have elementary skills in both word processing and electronic spreadsheet software, such as Office 365 or MS Office.

- Computers and WIFI facilities are extensively available for student use throughout KOI. Students are encouraged to make use of the campus Library for reference materials.
- Students will require access to the internet and email. Where students use their own computers, they should have internet access. KOI will provide access to required software.

Resource requirements specific to this subject: ASP.NET, Office 365, MS Imagine, Visual Studio - 2019, MS Azure, GitHub Desktop.

1.8 Academic Advising

Academic advising is available to students throughout teaching periods including the exam weeks. As well as requesting help during scheduled class times, students have the following options:

- Consultation times: A list of consultation hours is provided on the homepage of Moodle where appointments can be booked.
- Subject coordinator: Subject coordinators are available for contact via email. The email address of the subject coordinator is provided at the top of this subject outline.
- Academic staff: Lecturers and Tutors provide their contact details in Moodle for the specific subject. In most cases, this will be via email. Some subjects may also provide a discussion forum where questions can be raised.
- Head of Program: The Head of Program is available to all students in the program if they need advice about their studies and KOI procedures.
- Vice President (Academic): The Vice President (Academic) will assist students to resolve complex issues (but may refer students to the relevant lecturers for detailed academic advice).

2 Academic Details

2.1 Overview of the Subject

The subject focuses on advanced web design and development for small and large-scale applications. It gives the student a solid grounding in high-level programming language and allows them to build dynamic web applications using advanced internet technologies. The students will design, implement, and evaluate anything from small, personal websites to large-class web applications using a widely used framework in the industry. Students will apply object-oriented concepts to real-world problems, writing syntax code, creating SQL databases, developing interactive websites and composing user-friendly web applications for real-world problems.

2.2 Graduate Attributes for Undergraduate Courses

Graduates of Bachelor courses from King's Own Institute (KOI) will achieve the graduate attributes expected under the Australian Qualifications Framework (2nd edition, January 2013). Graduates at this level will be able to apply a broad and coherent body of knowledge from their major area of study in a range of contexts for professional practice or scholarship and as a pathway for further learning.

King's Own Institute's generic graduate attributes for a bachelor's level degree are summarised below:

	KOI Bachelor Degree Graduate Attributes	Detailed Description
	Knowledge	Current, comprehensive, and coherent and connected knowledge

	Critical Thinking	Critical thinking and creative skills to analyse and synthesise information and evaluate new problems
	Communication	Communication skills for effective reading, writing, listening and presenting in varied modes and contexts and for transferring knowledge and skills to a variety of audiences
	Information Literacy	Information and technological skills for accessing, evaluating, managing and using information professionally
	Problem Solving Skills	Skills to apply logical and creative thinking to solve problems and evaluate solutions
	Ethical and Cultural Sensitivity	Appreciation of ethical principles, cultural sensitivity and social responsibility, both personally and professionally
	Teamwork	Leadership and teamwork skills to collaborate, inspire colleagues and manage responsibly with positive results
	Professional Skills	Professional skills to exercise judgement in planning, problem solving and decision making

Across the course, these skills are developed progressively at three levels:

- **Level 1 Foundation** – Students learn the basic skills, theories and techniques of the subject and apply them in basic, standalone contexts
- **Level 2 Intermediate** – Students further develop the skills, theories and techniques of the subject and apply them in more complex contexts, and begin to integrate this application with other subjects.
- **Level 3 Advanced** – Students demonstrate an ability to plan, research and apply the skills, theories and techniques of the subject in complex situations, integrating the subject content with a range of other subject disciplines within the context of the course.

2.3 Subject Learning Outcomes

This is a Level 3 subject.

On successful completion of this subject, students should be able to:

Subject Learning Outcomes	Contribution to Graduate Attributes
a) Compose advanced solutions using object-oriented programming for problem-solving	   
b) Implement and evaluate dynamic web applications to meet design principles and user requirements	   
c) Develop dynamic small to large-class web applications with SQL database	   
d) Design and validate advanced user interfaces using creative programming packages served as the engine through data annotations	   



2.4 Subject Content and Structure

Below are details of the subject content and how it is structured, including specific topics covered in lectures and tutorials. Reading refers to the text unless otherwise indicated.

Weekly Planner:

Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
Week 1 02 March	Getting started with C# programming language	Ch.1 <i>Programming C# 12</i>	Install and explore visual studio. Solve C# exercises to get started with the programming language.
Week 2 09 March	C# Control statements, Arrays and Collections	Ch. 2 <i>Programming C# 12</i>	Solve C# exercises on control statements, arrays and lists. Biweekly activity 0%
Week 3 16 March	Object oriented programming (Classes)	Ch. 3 & 6 <i>Programming C# 12</i>	Solve C# exercises on object-oriented classes, attributes and methods.
Week 4 23 March	Introduction to web development and ASP.Net MVC5	Ch. 1 <i>Modern Web Development</i>	Solve exercises to create Asp.net MVC application. Biweekly activity 5% Quiz 1-10% (In class)
Week 5 30 March	Controllers Ethical web development	Ch. 4 <i>Modern Web Development</i>	Solve exercises to create Asp.net controllers. ACS New Code of Ethics
Week 6 07 April (Tue)	View and Razor Programming	Ch. 5 & 6 <i>Modern Web Development</i>	Solve exercises to create MVC views. Biweekly activity 5%
Week 7 13 April	Creating Model Component	Ch. 6 <i>Modern Web Development</i>	Solve exercises to create SQL server database and MVC models.
Week 8 20 April	Advanced Forms and HTML	Ch. 7 <i>Modern Web Development</i>	Solve exercises to create MVC forms. Biweekly activity 5% Quiz 2 -10% (In class)
Week 9 28 April (Tue)	SQL Complex Entity Relationship	Ch. 6 <i>Modern Web Development</i>	Solve exercises to create complex entity relationships.



Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
Week 10 04 May	Data Annotations Attributes in Depth	Ch. 6 <i>Modern Web Development</i>	Solve exercises to validate MVC forms using data annotation. Biweekly activity 5%
Week 11 11 May	Web Security and data privacy	Ch. 11 Modern Web Development	Solve exercises for web security.
Week 12 18 May	Testing	Ch. 13 <i>Modern Web Development</i>	Solve exercises based on testing. Assignment 3A due Assignment 3B due
Week 13 25 May	Study review week and Final Exam Week		
Week 14 01 June	Examinations Continuing students - enrolments for T226 open		Please see exam timetable for exam date, time and location
Week 15 09 June (Tue)	Student Vacation begins New students - enrolments for T226 open		
Week 16 15 June	- Results Released - Review of Grade Day for T126 – see Sections 2.6 and 3.2 below for relevant information. - Certification of Grades NOTE: More information about the dates will be provided at a later date through Moodle/KOI email.		
T226 29 June 2026			
Week 1 29 June	Week 1 of classes for T226		

2.5 Public Holiday Amendments

Please note: KOI is closed on all scheduled NSW Public Holidays.

T126 has Five (5) public holidays that occur during this trimester. Classes scheduled for these public holidays (Calendar Class Date) will be rescheduled as per the table below.

This applies to ALL subjects taught in T126.

Please see the table below and adjust your class timing as required. Please make sure you have arrangements in place to attend the rescheduled classes if applicable to your T126 enrolment.

Classes will be conducted at the same time and in the same location as your normally scheduled class except these classes will be held on the date shown below.



Calendar Class Date	Rescheduled Class Date
03 April 2026 (Friday) 04 April 2026 (Saturday) 06 April 2026 (Monday) 25 April 2026 (Saturday) 27 April 2026 (Monday)	25 May 2026 (Monday) 26 May 2026 (Tuesday) 27 May 2026 (Wednesday)

2.6 Review of Grade, Deferred Exams & Supplementary Exams/Assessments

Review of Grade:

There may be instances when you believe that your final grade in a subject does not accurately reflect your performance against the marking criteria. Section 8 of the *Assessment and Assessment Appeals Policy* (www.koi.edu.au) describes the grounds on which you may apply for a Review of Grade.

If you have a concern about your marks and you are unable to resolve it with the Academic staff concerned, then you can apply for a formal Review of Grade as explained in section 3.2(e) Appeals Process below. Please note the time limits for requesting a review. Please ensure you read the Review of Grade information before submitting an application.

Review of Grade Day:

Final exam scripts will not normally be returned to students. Students can obtain feedback on their exam performance and their results for the whole subject at the Review of Grade Day. KOI will hold the Review of Grade Day for all subjects studied in T126. **The ROG day will be announced at a later date and the students will be notified through Moodle/KOI email.**

Only final exams and whole subject results will be discussed as all other assessments should have been reviewed during the trimester. Further information about Review of Grade Day will be available through Moodle.

If you fail one or more subjects and you wish to consider applying for a Review of Grade you are **STRONGLY ADVISED to** attend the Review of Grade Day. You will have the chance to discuss your final exam and subject result with your lecturer, and will be advised if you have valid reasons for applying for a Review of Grade (see Section 3.2 below and the *Assessment and Assessment Appeals Policy*).

A formal request for a review of grade may not be considered unless you first contact the subject coordinator to discuss the result.

Deferred Exams:

If you wish to apply for a deferred exam because you are unable to attend the scheduled exam, you should submit the Assignment Extension / Exam Deferment Form available by clicking the following link [Assignment Extension / Exam Deferment Form](#) as soon as possible, *but no later than three (3) working days of the assessment due date*.

If you miss your final exam there is no guarantee you will be offered a deferred exam.

You must apply within the stated timeframe and satisfy the conditions for approval to be offered a deferred exam (see Section 8.1 of the *Assessment and Assessment Appeals Policy and the Application for Assignment Extension or Deferred Exam Forms*). In assessing your request for a deferred exam, KOI will take into account the information you provide, the severity of the event or circumstance, your performance on other items of assessment in the subject, class attendance and your history of previous applications for special consideration.

Deferred Quiz 1 and Quiz 2 will be held before the end of week 9. Deferred final exams will be held before next trimester. You will not normally be granted a deferred exam on the grounds that you mistook the time, date or place of an examination, or that you have made arrangements to be elsewhere at that time; for example, have booked plane tickets.



If you are offered a deferred exam, but do not attend *you will be awarded 0 marks for the exam*. This may mean it becomes difficult for you to pass the subject. If you apply for a deferred exam within the required time frame and satisfy the conditions you will be advised by email (to your KOI student email address) of the time and date for the deferred exam. Please ensure that you are available to take the exam at this time.

Marks awarded for the deferred exam will be the marks awarded for that item of assessment towards your final mark in the subject.

Supplementary Assessments (Exams and Assessments):

A supplementary assessment may be offered to students to provide a final opportunity to demonstrate successful achievement of the learning outcomes of a subject. *Supplementary assessments are only offered at the discretion of the Board of Examiners*. In considering whether or not to offer a supplementary assessment, KOI will take into account your performance on all the major assessment items in the subject, your attendance, participation and your history of any previous special considerations.

If you are offered a supplementary assessment, you will be advised by email to your *KOI student email address* of the time and due date for the supplementary assessment – supplementary exams will normally be held at the same time as deferred final exams during week 1 or week 2 of the next trimester.

You must pass the supplementary assessment to pass the subject. The maximum grade you can achieve in a subject based on a supplementary assessment is a PASS grade.

If you:

- are offered a supplementary assessment, but fail it;
- are offered a supplementary exam, but do not attend; or
- are offered a supplementary assessment but do not submit by the due date; you will receive a FAIL grade for the subject.

Students are also eligible for a supplementary assessment for their final subject in a course where they fail the subject but have successfully completed all other subjects in the course. You must have completed all major assessment tasks for the subject and obtained a passing mark on at least one of the major assessment tasks to be eligible for a supplementary assessment.

If you believe you meet the criteria for a supplementary assessment for the final subject in your course, but have not received an offer, complete the *Complaint, Grievance, Appeal Form* and send your form to reception@koi.edu.au. The deadline for applying for supplementary assessment is the Friday of the first week of classes in the next trimester.

2.7 Teaching Methods/Strategies

Briefly described below are the teaching methods/strategies used in this subject:

- *Lectures* (2 hours/week) are conducted in seminar style and address the subject content, provide motivation and context and draw on the students' experience and preparatory reading.
- *Tutorials* (2 hours/week) include class discussion of case studies and research papers, practice sets and problem-solving and syndicate work on group projects. Tutorials often include group exercises and so contribute to the development of teamwork skills and cultural understanding. Tutorial participation is an essential component of the subject and contributes to the development of many of the graduate attributes (see section 2.2 above). Tutorial participation contributes towards the assessment in many subjects (see details in Section 3.1 for this subject). Supplementary tutorial material such as case studies, recommended readings, review questions etc. will be made available each week in Moodle.
- *Online* teaching resources include class materials, readings, model answers to assignments and exercises and discussion boards. All online materials for this subject as provided by KOI will be found in the Moodle page for this subject. Students should access Moodle regularly as material may be updated at any time during the trimester



- **Other contact** - academic staff may also contact students either via Moodle messaging, or via email to the email address provided to KOI on enrolment.

2.8 Student Assessment

Assessment is designed to encourage effective student learning and enable students to develop and demonstrate the skills and knowledge identified in the subject learning outcomes. Assessment tasks during the first half of the study period are usually intended to maximise the developmental function of assessment (formative assessment). These assessment tasks include weekly tutorial exercises (as indicated in the weekly planner) and low stakes graded assessment (as shown in the graded assessment table). The major assessment tasks where students demonstrate their knowledge and skills (summative assessment) generally occur later in the study period. These are the major graded assessment items shown in the graded assessment table.

Final grades are awarded by the Board of Examiners in accordance with KOI's Assessment and Assessment Appeals Policy. The definitions and guidelines for the awarding of final grades within the BIT degree are:

- **HD High distinction (85-100%)** an outstanding level of achievement in relation to the assessment process.
- **DI Distinction (75-84%)** a high level of achievement in relation to the assessment process.
- **CR Credit (65-74%)** a better than satisfactory level of achievement in relation to the assessment process.
- **P Pass (50-64%)** a satisfactory level of achievement in relation to the assessment process.
- **F Fail (0-49%)** an unsatisfactory level of achievement in relation to the assessment process.

Provided below is a schedule of formal assessment tasks and major examinations for the subject.

Assessment Type	When assessed	Weighting	Learning Outcomes Assessed
Assignment 1 – Biweekly (2,4,6,8 and 10) Individual Progress Reports and Reflection - Individual Assignment	Week 2- 0% Week 4- 5% Week 6- 5% Week 8 - 5% Week 10 - 5%	20%	a, b, c, d
Assignment 2: Quiz 1 and Quiz 2 (In Class)	Week 4-10% Week 8-10%	20%	a, b, c, d
Assignment 3A: MVC web application - Group Assignment	Week 12	35%	b, c, d
Assignment 3B: MVC web application Group report and video presentation	Week 12	25%	b, c, d

Requirements to Pass the Subject:

To gain a pass or better in this subject, students must gain a *minimum of 50%* of the total available subject marks.



2.9 Prescribed and Recommended Readings

Provided below, in formal reference format, is a list of the prescribed and recommended readings.

Prescribed Texts:

Griffiths, I., 2024, *Programming C# 12: Build Cloud, Web, and Desktop Applications*, O'Reilly Media, Inc., Sebastopol, CA. ISBN: 9781098158361.

Peres, R., 2020, *Modern Web Development with ASP.NET Core 3*, 2nd ed., Packt Publishing, Birmingham, UK. ISBN: 9781789614282.

Griffiths, I., 2022, *Programming C# 10: Build Cloud, Web, and Desktop Applications*, O'Reilly Media, Inc., Sebastopol, CA. ISBN: 9781098117818.

Recommended Readings:

Mezei, R.A., 2023, *Introduction to the Development of Web Applications Using ASP.NET (Core) MVC*, 1st ed., Springer Nature, Cham, Switzerland. ISBN: 9783031306259.

Price, M.J., 2024, *C# 13 and .NET 9 – Modern Cross-Platform Development Fundamentals*, 9th ed., Packt Publishing, Birmingham, UK. ISBN: 9781835881224.

Frain, B., 2022, *Responsive Web Design with HTML5 and CSS*, 4th ed., Packt Publishing, Birmingham, UK. ISBN: 9781803242712.

Rebah, H.B., Boukthir, H., and Chedebois, A., 2022, *Website Design and Development with HTML5 and CSS3*, John Wiley & Sons, Hoboken, NJ. ISBN: 9781786306968.

Journal and conference articles

Akyon, S.H., Akyon, F.C., and Yilmaz, T.E., 2023, Artificial intelligence-supported web application design and development for reducing polypharmacy side effects and supporting rational drug use in geriatric patients, *Frontiers in Medicine*, vol. 10, article 1029198. DOI: 10.3389/fmed.2023.1029198.

Khan, S., and Khanam, A.T., 2023, Study on MVC framework for web development in PHP, *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, pp. 414–419. DOI: 10.32628/CSEIT2390450.

Landgraf, G., 2021, How user-friendly is your website? Usability lessons for libraries in a remote world, *American Libraries*, vol. 52, no. 3/4, pp. 30–33. Available at:
<https://search.ebscohost.com/login.aspx?direct=true&db=eue&AN=149005478&site=ehost-live>

Negri-Ribalta, C., Lombard-Platet, M., and Salinesi, C., 2024, Understanding the GDPR from a requirements engineering perspective - a systematic mapping study on regulatory data protection requirements', *Requirements Engineering*, pp. 1–27. DOI: 10.1007/s00766-024-00423-4.



Huynh, T., 2024, *Web technologies comparison between Nuxt.js and ASP.NET*, Thesis/Technical Report, Centria University of Applied Sciences. <https://www.theseus.fi/handle/10024/818165>

Klug, C., 2023, *Advanced ASP.NET Core*, *Visual Studio Magazine*, 14 September. Available at: <https://visualstudiomagazine.com/articles/2023/09/14/advanced-aspnet-core.aspx>

Australian Computer Society (ACS), 2024, *The ACS New Code of Ethics Released*, Member Insights, viewed 17 February 2026. Available at: <https://membership.acs.org.au/member-insight/20240117-ACS-New-Code-of-Ethics-Released.html>

Students are encouraged to read peer reviewed journal articles and conference papers. Google Scholar provides a simple way to broadly search for scholarly literature. From one place, you can search across many disciplines and sources: articles, theses, books, abstracts and court opinions, from academic publishers, professional societies, online repositories, universities and other web sites.

Useful Websites:

The following industry websites are useful introductory sources covering a range of information useful for this subject.

- <https://learn.microsoft.com/en-us/aspnet/mvc/overview/getting-started/introduction/getting-started>
- <https://www.tutorialsteacher.com/mvc>
- https://www.w3schools.com/asp/webpages_intro.asp
- <https://www.geeksforgeeks.org/mvc-design-pattern/>

3. Assessment Details

3.1 Details of Each Assessment Item

The assessments for this subject are described below. Other assessment information and/or assistance can be found in Moodle. Marking guides for assessments follow the assessment descriptions. Students should compare final drafts of their assessment against the marking guide before submission.

Assessment submission is as per the instructions below. Please retain a photocopy and softcopy of all assessments.

KOI expects students to submit their own original work in both assignments and exams, or the original work of their group in the case of group assignments.

Assessment 1

Assessment type: Biweekly activities - Individual Assignment (20%), Individual Progress Report and Reflection (300-500 words)

Purpose: Biweekly reports to evaluate individual progress. The students must address web development issues/problems that they may face in the industry. Biweekly activities will be individual activities and may be related to code design patterns, peer evaluations of developed programs, critical thinking of existing systems, etc. This assessment contributes to learning outcomes a, b, c, and d.

Value: 20% **Due Date:** Biweekly submission

Submission requirements details: Week 2 (0%) + Week 4 (5%) + Week 6 (5%) + Week 8 (5%) + Week 10 (5%) = Total (20%)



Assessment topic: Progress Report and Reflection on two weeks in class activities

Task Details: The requirements for each biweekly assessment will be provided on Moodle. Students will be expected to prepare the report accordingly including screen shots.

Week 2 activity:

It is required to identify one job from job search websites such as seek.com.au which relates to ASP.net development, C# development, and MVC development.

Complete the template below:

JOB SEARCH WEBSITE:
COMPANY:
POST DATE:
MAIN TASKS:
SKILLS REQUIRED:

Week 4 activity:

Critical thinking skill is essential to learn problem-solving. It is used to understand the logical connections between ideas. Your tutor will provide you with a C# console program. You need to do the following tasks:

- Task 1: Download and explore the code using Visual Studio 2019 or 2022. Add a screenshot of the code from Visual Studio and include it in the report.
- Task 2: Describe the problem that the coder is solving in the program.
- Task 3: Consider redesigning and recoding the program by identifying four changes in the C# code using functions, classes, lists, etc. For each change, you need to display a code snippet screenshot before the change and another after the changes, accompanied by a description of what you modified.

Submission: Create a pdf document or word document to address tasks 1-3. Support your answers with screenshots.

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Structure and Format Value 1% Mark awarded	Very difficult to Read, unclear structure and sections missing.	Some difficulty in reading, not very clear, important sections included.	Clear and readable and all required sections are included.	Well written and very clear, with all required sections with completed discussions included.	Very Well written and very clear. All sections are included and discussed, and Additional information is included for clarity.
Problem identification Value 4% Mark awarded	No attempt for the tasks	Attempt one task	Attempt two tasks	Attempt three tasks	Attempt all the tasks



Total Marks: / 5%	COMMENTS:
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Week 6 activity:

GitHub is an open-source version control used for software development. Software development companies prefer developers with GitHub experience as teams develop projects. This activity requires working on a program that you developed in any previous lab. You need to do the following tasks:

- Task 1: Create a GitHub account (<https://github.com/>).
- Task 2: Download the GitHub desktop (<https://desktop.github.com/>).
- Task 3: Upload any program from the previous lab to the GitHub account using the GitHub desktop.
- Task 4: Make two changes to the code and check the files' statuses.
- Task 5: Change some of the files and upload them to GitHub account.
- Task 6: Discuss what you have learned from this activity.

Submission: You must create and submit a video presentation of 10-15 mins to explain Tasks 1-6. If the video file is large, you can upload it to google drive and submit the shared link.

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Video format 1% Mark awarded	No video presentation or major issues with video and audio.	Some difficulty in understanding the outcomes.	Clear audio, but the student doesn't show the presentation slides.	Good video quality	Outstanding video quality
Problem identification Value 4% Mark awarded	No attempt for the tasks	Attempt less than two tasks	Attempt less than three tasks	Attempt less than four tasks	Attempt all the tasks
Total Marks: / 5%	COMMENTS:				

Week 8 activity: Research on ACS New Code of Ethics and its Application in Software Development

You are required to research the ACS New Code of Ethics and explore how software developers can align their practices with the principles outlined in the code. Focus on understanding the ethical responsibilities of developers, including professionalism, accountability, and the broader social impacts of technology.

Describe how the ACS Code of Ethics applies to various stages of software development, from design to deployment. Highlight at least four key aspects of the software development process where ethical considerations, such as privacy, security, competence, and honesty, should be prioritized according to the ACS Code.

Submission: You must create and submit a video presentation of 5 mins to explain your results. If the video file is large, you can upload it to google drive and submit the shared link.



Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Video format 1% Mark awarded	No video presentation or major issues with video and audio.	Some difficulty in understanding the outcomes.	Clear audio, but the student doesn't show the presentation slides.	Good video quality	Outstanding video quality
Problem identification Value 2% Mark awarded	No attempt or the problem is not identified	Identify only soft or hard skills.	Identify less than four soft and hard skills	Identify sufficient soft and hard skills	Identify outstanding soft and hard skills
Improvement Value 2% Mark awarded	No attempt	The answer is unreasonable, with major flaws and unclear explanations.	An appropriate explanation for ways to improve the skills	Clear and detailed explanation to improve the skills	Accurate and Thorough explanation provided to improve the skills
Total Marks: / 5%	COMMENTS:				

Week 10 activity:

A web hosting service makes websites available for users on the World Wide Web (www). Companies provide web hosting based on several specifications. Identify one of the companies that offer web hosting services from the internet and discuss what is required to host your ASP.net web application in assessment 3. It is required to discuss and justify the selection of the following issues:

1. Web hosting types, including shared, VPS, and cloud hosting.
2. Server specification
3. Service level agreements such as uptime guarantee, response time, etc.
4. Support details
5. Price
6. Security tools

Submission: You must create and submit a video presentation of 5-7 mins to present

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Structure and Format Value 1% Mark awarded	Very difficult to Read, unclear structure and sections missing.	Some difficulty in reading, not very clear, important sections included.	Clear and readable and all required sections are included.	Well written and very clear, with all required sections with completed discussions included.	Very Well written and very clear. All sections are included and discussed, and Additional information is included for clarity.
Problem identification Value 4% Mark awarded	No attempt for the tasks	Attempt less than two tasks	Attempt less than three tasks	Attempt less than four tasks	Attempt all the tasks
Total Mark: / 5%	COMMENTS:				



Assessment 2

Assessment type: Quiz - Individual

Purpose: Students are required to answer a set of questions in quiz 1 covering topics 1,2 and 3 and quiz 2 covering topics 1-7. This assessment contributes to learning outcomes a, b, c and d. This assessment must be in class.

- Attend their enrolled class **in person** to do the quiz, unless their enrollment allows online attendance.

Value: 20% (Quiz 1 is 10% and Quiz 2 is 10%)

Due Date: Week 4 and Week 8 respectively.

Task Details: The quiz will consist of a series of short answer questions relating to subject content weeks 1 – 3 inclusive and weeks 4 - 7 inclusive respectively.

Assessment 3A

Assessment type: Create MVC web application – Group Assignment (3-4 students per group)

Assessment purpose: This assessment will allow students to demonstrate that they can develop dynamic websites using ASP.NET Data Controls. Students are required to develop, test and maintain a dynamic Internet application for business using an integrated suite of software tools. This assessment contributes to learning outcomes b, c, d and e.

Value: 35%

Due date: Week 12

Assessment topic: Create an MVC web application

Task Details: Library Management System

You have been selected to develop a web application for an online library management platform. This platform will allow library administrators to manage book collections and borrowing operations, while providing library members with the ability to browse, borrow, and manage their reading activities. Your task is to create a comprehensive library management system that serves both library administrators and library members.

Librarians / Library Administrators

- Should be able to create and manage library profiles, including adding library information such as name, location, operating hours, and contact details.
- Must have the ability to upload and manage book records, including titles, authors, genres, ISBN numbers, availability status, and book cover images.
- Should be able to configure borrowing parameters such as loan duration, renewal limits, overdue penalties, and maximum number of borrowable items.
- Must be able to track all borrowing transactions, view which books are borrowed or reserved, and manage return statuses and fines.
- Should be able to review and manage feedback or ratings submitted by members for books.

Library Members / Readers

- Should be able to browse available books, view book details including summaries, authors, genres, availability, and cover images.
- Must be able to borrow or reserve books online, view borrowing status (e.g., borrowed, reserved, overdue) and manage renewals where applicable.



- Should have access to their borrowing history, including previously borrowed books, due dates, fines paid, and feedback submitted.
- Must be able to leave feedback or rate books they have read.

System Features

- The system should include a secure login and registration process for both librarians and library members to ensure protection of personal and account information.
- A user-friendly homepage should be available for both librarians and members, displaying relevant information such as new arrivals, most borrowed books, recommendations, and currently available books.
- The system should generate detailed reports on library activity, including borrowing trends, overdue books, most active members, and most popular books.

Model Design Requirements:

- Use the MVC architecture to develop the application.
- Design your entity-relationship diagram (ERD) for the classes based on your understanding of complex entity relationships, including one-to-one, one-to-many, and many-to-many relationships.
- Use appropriate data annotation and input validations for the HTML elements in the MVC forms. You can either use custom or default error messages.

Application Requirements:

- The web application must be created using MVC ASP.NET technology using C# for the backend and SQL express for the database.
- The home page must show group details, including student ID and student full names, in a table.
- All the pages should have a navigation menu.
- Use CSS to create a custom design for the web pages.
- The web application must be ready to run without any additional configurations.

Submission requirements details:

All files must be uploaded on Moodle as a zipped File. If the file is large, you can upload it to google drive and submit the shared link with the group details.

For the individual task, students are required to demonstrate to the tutor the tasks they completed in the assessment during the weekly labs from lab 10 to 12. Additionally, the tutor may inquire about the methods used to accomplish these tasks. Each of these three weeks' tasks is worth 3%, contributing a total of 9% overall. Furthermore, a document related to these tasks needs to be submitted individually, and each week's submission is worth 2%.

Marking Rubric for Assessment 3A – Group Task:

Value 20%

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Use MVC architecture. Database is created Value 3% Mark awarded	MVC is not used. The database is not or partially created.	MVC is configured, but the database does not exist	MVC is configured, the database is created but not all the models and attributes are translated to tables and columns	MVC is configured, the database is created based on the ERD	MVC is configured, the database is created based on the ERD. Custom actions in the controllers are created



Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Application is running Value 5% Mark awarded	Webpages are not displayed on the web browser. The system is not running.	Some of the pages are displayed, but there are some errors in storing information using web pages.	All the webpages can be displayed without errors but there are some errors to store information in the database	Webpages are fully functional. Information can be created and stored in the database without errors.	Webpages are fully functional. Information can be created and stored in the database without errors. Web Pages are linked.
Create pages that allow adding new records Value 5% Mark awarded	No web pages in the application	Some web pages are created	All the webpages are analysed and some are created based on the model classes but they are not all evaluated properly	All the webpages are created and evaluated based on the model classes.	All the webpages are created and evaluated based on the model classes. Custom HTML elements are added into the webpages. Home page includes the group details.
Appropriate data annotation is used Value 2% Mark awarded	Not implemented in the webpages	Some webpages have validation rules using data annotation	Evaluate the validation rules for some webpages using data annotation	Create and evaluate validation rules for all webpages. The web pages are validated correctly and the error messages are displayed on the forms	Create and evaluate the validation rules for all the webpages using data annotation. The web pages are validated correctly and the error messages are displayed on the forms. Custom error messages are used
Web security Value 3% Mark awarded	Not Implemented	Create a partial or not working login portal to the system.	Create a proper login portal to the system.	Create a robust login portal to the system using advanced security rules.	Create and evaluate a robust login portal to the system using advanced security rules.
Web interface is designed professionally and attractive. CSS is used. Value 2%	Web design is not acceptable.	Web design is basic using the default MVC theme. Logo is added	Web design is fairly analysed. Logo is added. Some basic CSS rules are used.	Web design is attractive. Logo is added. Complex CSS rules are used.	Web design is fancy and professional
Total Marks: / 20%	COMMENTS:				

Marking Rubric for Assessment 3A – Individual Task: Value 15%

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Task demonstration 9% (3% for each week from 10-12) Mark awarded	Little to no evidence of task demonstration during weekly labs.	Demonstrates the completed tasks during weekly labs, though with some inaccuracies or errors.	Effectively demonstrates completed tasks during weekly labs with a good level of accuracy.	Demonstrates completed tasks with high accuracy and confidence during weekly labs.	Demonstrates completed tasks flawlessly and expertly during weekly labs.



Document Submission 6% (2% for each week from 10-12) Mark awarded	Fails to submit the required document(s) for any or all weeks.	Submits the required document, but may be incomplete or with errors.	Submits the required document with reasonable completeness and accuracy.	Submits the required document with a high level of completeness and accuracy.	Submits the required document flawlessly and with exceptional quality.
Total Marks: / 15%	COMMENTS:				

Assessment 3B

Assessment type: Group report and video presentation

Assessment purpose: To allow students to prepare a report and demonstrate their work completed in assignment 3A. This assessment contributes to learning outcomes b, c, d and e.

Value: 25%

Due date: Week 12

Assessment topic: Web application report and video presentation

Task details:

Report:

- Prepare a report on the website project created in Assignment 3A
- The report should be appropriately structured and formatted to show the following sections:
 - Introduction and background about the system. What are the functional requirements and benefits of the system?
 - Explain the tasks that you achieved in the application
 - Add the entity-relationship diagram into the report and justify why or how the proposed architecture is suitable for the case study.
 - Use the literature review to discuss the testing strategy, including Unit testing, acceptance testing, etc. for the web application.
 - Add a user manual to the web application.

Individual Video presentation:

- Create a 3 minute individual video presentation.
- In the video presentation:
 - Demonstrate the tasks that you have achieved in the group

Submission requirements details: The report must be submitted as a pdf or word document.

Marking Rubric for Assessment 3B: Value 25%

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Structure and Format Value 3% Mark awarded	Very difficult to Read, unclear structure and sections missing.	Some difficulty in reading, not very clear, important sections included.	Clear and readable and all required sections are included.	Well written and very clear, with all required sections with completed discussions included.	Very Well written and very clear. All sections are included and discussed, and Additional information is



					included for clarity.
Introduction and background Value 4% Mark awarded	No introduction was given or was irrelevant.	The introduction is provided with some details and limited cohesion.	Objectives are analysed. The introduction is provided with most of the required details in a cohesive manner.	Objectives are analysed. Introduction is created with all the required details comprehensively and cohesively.	Objectives are analysed. Introduction is created, and all details are presented in a systematically comprehensive and cohesive manner.
ERD justification Value 2% Mark awarded	Justification is Incorrect, Irrelevant or missing.	The answer is unreasonable with major flaws and unclear explanation.	Appropriate justification provided for ERD.	Clear and detailed justification provided for ERD.	Accurate and Thorough explanation provided for all classes used in the ERD.
Security analysis Value 3%	Not or partially covered.	Security aspects are partially analysed.	Security aspects are fully analysed and justified using literature.	Security aspects are fully analysed and justified using literature.	Advanced Security aspects are fully analysed and justified using literature.
User manual. Value 3%	The user manual is not added.	The user manual is very basic without screenshots.	User manual is acceptable. With some screenshots.	The user manual is professional.	The user manual is professional based on some standards.
Video presentation Value 10%	Unclear communication, lacking understanding of the work done.	Some information was communicated with little understanding of the work.	Appropriate communication with a clear understanding of the work.	Good communication and delivery, with a detailed explanation of the completed work.	Excellent presentation with additional visual aids and a thorough understanding of concepts and the techniques used in work.
Total Marks: / 25%	COMMENTS:				

3.2 General information about assessment

a) Late Penalties and Extensions

An important part of business life and key to achieving KOI's graduate outcome of Professional Skills is the ability to manage workloads and meet deadlines. Completing assessment tasks on time is a good way to master these habits.

Students who miss Quizzes and final exams without a valid and accepted reason may not be granted a deferred exam and will be awarded 0 marks for the assessment item. Assessment items which are missed or submitted after the due date/time will attract a penalty unless there is a compelling reason (see below). These penalties are designed to encourage students to develop good time management practices, and to create equity for all students.

Any penalties applied will only be up to the maximum marks available for the specific piece of assessment attracting the penalty.

Late penalties, granting of extensions and deferred exams are based on the following:



In Class Tests and Quizzes

- Generally, extensions are not permitted. A make-up test may only be permitted under very special circumstances where acceptable supporting evidence of illness, hardship or unavoidable problems preventing completion of the assessment is provided (see section (b) below). The procedures and timing to apply for a make-up test (only if available) are as shown in the section *Applying for an Extension* (see below).
- Missing a class test will result in 0 marks for that assessment item unless the above applies.

Written Assessments and Video Assessments

- There is a late penalty of *5% of the total available marks* per calendar day unless an extension is approved (see *Applying for an Extension* section below).

Presentations

- Generally, extensions are not permitted. Missing a presentation will result in 0 marks for that assessment item. The rules for make-up presentations are the same as for missing in-class tests (described above).
For group presentations, if serious circumstances prevent some members of the group from participating, the members of the group who are present should make their contributions as agreed. If a make-up presentation is approved, the other members of the group will be able to make their individual presentation later and will be marked according to the marking rubric. A video presentation may be used to facilitate the process.

Final Exams

If students are unable to attend final exams due to illness, hardship or some other unavoidable problem (acceptable to KOI), they must:

- Complete the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible, *but no later than three (3) working days after the exam date*.
- Provide acceptable documentary evidence (see section (b) below).
- Agree to attend the deferred exam as set by KOI if a deferred exam is approved.

Deferred exam

- There will only be one deferred exam offered.
- Marks obtained for the deferred exam will be the marks awarded for that assessment item.
- If you miss the deferred exam you will be awarded 0 marks for the assessment item. This may mean you are unable to pass the subject.

b) Applying for an Extension

If students are unable to submit or attend an assessment when due, they must

- Complete the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible, *but no later than three (3) working days of the assessment due date*.
- Provide acceptable documentary evidence in the form of a medical certificate, police report or some other appropriate evidence of illness or hardship, or a technician's report on problems with



computer or communications technology, or a signed and witnessed statutory declaration explaining the circumstances.

- Students and lecturers / tutors will be advised of the outcome of the extension request as soon as practicable.

Please remember there is no guarantee of an extension being granted, and poor organisation is not a satisfactory reason to be granted an extension.

c) Referencing and Plagiarism

Please remember that all sources used in assessment tasks must be suitably referenced.

Failure to acknowledge sources is plagiarism, and as such is a very serious academic issue. Students plagiarising run the risk of severe penalties ranging from a reduction in marks through to 0 marks for a first offence for a single assessment task, to exclusion from KOI in the most serious repeat cases. Exclusion has serious visa implications. The easiest way to avoid plagiarising is to reference all sources.

Harvard referencing is the required method – in-text referencing using Author's Surname (family name) and year of publication. A Referencing Guide, "Harvard Referencing", and a Referencing Tutorial can be found on the right-hand menu strip in Moodle on all subject pages.

An effective way to reference correctly is to use *Microsoft Word's* referencing function (please note that other versions and programs are likely to be different). To use the referencing function, click on the References Tab in the menu ribbon – students should choose *Harvard*.

Authorship is also an issue under plagiarism – KOI expects students to submit their own original work in both assessment and exams, or the original work of their group in the case of a group project. All students agree to a statement of authorship when submitting assessments online via Moodle, stating that the work submitted is their own original work.

The following are examples of academic misconduct and can attract severe penalties:

- Handing in work created by someone else (without acknowledgement), whether copied from another student, written by someone else, or from any published or electronic source, is fraud, and falls under the general Plagiarism guidelines.
- Copying / cheating in tests and exams is academic misconduct. Such incidents will be treated just as seriously as other forms of plagiarism.
- Students who willingly allow another student to copy their work in any assessment may be considered to assisting in copying/cheating, and similar penalties may be applied.

Where a subject coordinator considers that a student might have engaged in academic misconduct, KOI may require the student to undertake an additional oral exam as a part of the assessment for the subject, as a way of testing the student's understanding of their work.

Further information can be found on the KOI website.

d) Reasonable Adjustment

The Commonwealth Disability Discrimination Act (1992) makes it unlawful to treat people with a disability less fairly than people without a disability. In the context of this subject, the principle of Reasonable Adjustment is applied to ensure that participants with a disability have equitable access to all aspects of the learning for the subject. For assessment, this means that barriers to their demonstrating competence are removed wherever it is reasonably practical to do so.

Examples of reasonable adjustment in assessment may include:

- provision of an oral assessment, rather than a written assessment
- provision of extra time
- use of adaptive technology.



The focus of the adjusted assessment should be on enabling the student to demonstrate achievement of the learning outcomes for the subject, rather than on the method of assessment.

e) Appeals Process

Full details of the KOI *Assessment and Assessment Appeals Policy* may be obtained in hard copy from the Library, and on the KOI website www.koi.edu.au under *Policies and Forms*.

Assessments:

Where students are not satisfied with the results of an assessment, they have the right to appeal. The process is as follows:

- Discuss the assessment with their tutor or lecturer – students should identify where they feel more marks should have been awarded – students should provide valid reasons based on the marking guide provided for the assessment. Reasons such as “*I worked really hard*” are not considered valid.
- If still not satisfied, students should complete an Application for Review of Assessment Marks form, clearly explaining the reasons for seeking a review. This form is available from the KOI website under *Policies and Forms* and is also available at KOI Reception (Kent St, Market St and O’Connell St). The completed Application for Review of Assessment Marks form should be submitted as explained on the form with supporting evidence attached to academic@koi.edu.au.
- The form must be submitted within *ten (10) working days* of the return of the marked assessment, or within *five (5) working days* after the return of the assessment if the assessment is returned after the end of the trimester.

Review of Grade – whole of subject and final exams:

Where students are not satisfied with the results of the whole subject or with their final exam results, they have the right to request a Review of Grade – see the *Assessment and Assessment Appeals Policy* for more information.

An *Application for Review of Grade/Assessment Form* (available from the KOI Website under *Policies and Forms* and from KOI Reception at Kent St, Market St and O’Connell St) should be completed clearly explaining the grounds for the application. The completed application should be submitted as explained on the form, with supporting evidence attached to academic@koi.edu.au.